Free pythagorean theorem worksheet worksheet 11

Maths | 9th grade | Mixed Operations

1. If 3x - 5 = 7, what is the value of x?

A) 2
B) 4
C) 5
D) 3

2. Solve for y in the equation: 2y + 3(4 - y) = 10.

A) 4
B) 2
C) 3
D) 5

3. The sum of three consecutive integers is 48. What is the smallest of these integers?

A) 15
B) 16
C) 14
D) 13

4. What is the value of x if 5(x - 2) = 3x + 4?

A) 2
B) 6
C) 4
D) 10

5. If the product of two numbers is 56 and one of the numbers is 7, what is the other number?

A) 8
B) 7
C) 6
D) 9

6. A rectangle has a length that is twice its width. If the perimeter is 36, what is the length?

A) 8
B) 12
C) 16
D) 10

7. Evaluate the expression: 4 + 3 * (8 - 6) / 2.

A) 6
B) 8
C) 10
D) 7

8. Find the value of y if 4(y - 3) = 2y + 8.

A) 7

B) 8
C) 5
D) 6

9. If the average of three numbers is 15 and two of the numbers are 12 and 18, what is the third number?

A) 14
B) 15
C) 16
D) 17

10. What is the solution to the inequality $5x - 7 > 3$?

A) $x > 2$
B) $x < 2$
C) $x > 1$
D) $x < 1$

Answer Key

1. A
Add 5 to both sides to get 3x = 12, then divide by 3 to find x = 4.

2. B
Distribute 3 to get 2y + 12 - 3y = 10. Simplify to -y + 12 = 10, then y = 2.

3. C
Let the smallest integer be x. Then the integers are x, x+1, x+2. Solve x + (x+1) + (x+2) = 48 to find x = 15.

4. B
Distribute 5 to get 5x - 10 = 3x + 4. Solve for x by subtracting 3x and adding 10 to both sides to get x = 7.

5. A
Divide 56 by 7 to find the other number, which is 8.

6. B
Let width = w, then length = 2w. Perimeter is 2(w + 2w) = 36. Solve for w = 6, so length = 12.

7. C
Calculate inside the parentheses: 8 - 6 = 2, then multiplication and division: 3 * 2 = 6, 6 / 2 = 3, finally addition: 4 + 3 = 7.

8. A
Distribute 4 to get 4y - 12 = 2y + 8. Subtract 2y and add 12 from both sides to solve for y = 10.

9. B
The sum of the numbers is 3 * 15 = 45. The third number is 45 - (12 + 18) = 15.

10. A
Add 7 to both sides to get 5x > 10, then divide by 5 to find x > 2.